Data Quality Audit

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Loading packages

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.5.3
```

```
library(tidyverse)
```

Import dataset

```
All_Country <- read_csv("Data/Processed/AllCountries_Processed.csv")
```

Dataset Structure & Summary

```
dim(All_Country)
```

```
## [1] 19449    25
```

```
summary(All_Country)
```

```
##    Country             HH_ID           Subdivision1       Subdivision2
##  Length:19449       Length:19449       Length:19449       Length:19449
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##   UrbanOrRural   ElectrificationStatus HouseholdSize     DwellingType
##  Min.   :1.000   Length:19449          Min.   : 1.000   Min.   :  1.000
##  1st Qu.:1.000   Class :character      1st Qu.: 3.000   1st Qu.:  1.000
##  Median :2.000   Mode  :character      Median : 4.000   Median :  1.000
##  Mean   :1.508                         Mean   : 4.777   Mean   :  2.675
##  3rd Qu.:2.000                         3rd Qu.: 6.000   3rd Qu.:  2.000
```

```
##  Max.   :3.000                          Max.   :23.000   Max.   :555.000
##  NA's   :3297                           NA's   :7119     NA's   :292
##  DwellingOwnership DwellingRoomsNumber  BankAccount    GridConnected
##  Min.   :1.000     Min.   :   0.000    Min.   :  1.0   Min.   :  0.000
##  1st Qu.:1.000     1st Qu.:   2.000    1st Qu.:  1.0   1st Qu.:  1.000
##  Median :1.000     Median :   3.000    Median :  2.0   Median :  2.000
##  Mean   :1.278     Mean   :   3.364    Mean   : 23.4   Mean   :  1.859
##  3rd Qu.:2.000     3rd Qu.:   4.000    3rd Qu.:  2.0   3rd Qu.:  2.000
##  Max.   :2.000     Max.   :2000.000    Max.   :111.0   Max.   :900.000
##  NA's   :289       NA's   :3925        NA's   :287     NA's   :3810
##  NumYearsGridConnected GridConnectionFeeLocal GridConnectionFeeUSD
##  Min.   :     0.0      Min.   :    -8         Min.   :  -0.6838
##  1st Qu.:     2.0      1st Qu.:     0         1st Qu.:   0.0000
##  Median :     4.0      Median :  1000         Median :  14.7059
##  Mean   :   312.1      Mean   : 25249         Mean   :  76.2559
##  3rd Qu.:    10.0      3rd Qu.: 35000         3rd Qu.:  70.4574
##  Max.   :100000.0      Max.   :800000         Max.   :6862.7451
##  NA's   :12616         NA's   :14453          NA's   :14453
##  InternalWiringFeeLocal InternalWiringFeeUSD ElectricityBillPriceLocal
##  Min.   :    -8         Min.   :  -0.6838    Min.   :   -8
##  1st Qu.:  4500         1st Qu.:  18.5414    1st Qu.:  500
##  Median : 12000         Median :  37.0828    Median : 1500
##  Mean   : 21943         Mean   :  68.9804    Mean   : 3626
##  3rd Qu.: 25250         3rd Qu.:  75.4717    3rd Qu.: 4000
##  Max.   :700000         Max.   :1698.5138    Max.   :99000
##  NA's   :16765          NA's   :16765        NA's   :14129
##  ElectricityBillPriceUSD ElectricityConsumptionkWh OutageDurationHours
##  Min.   : -0.6838        Min.   :   -8.00          Min.   :  -8.0
##  1st Qu.:  2.8302        1st Qu.:    1.00          1st Qu.:   2.0
##  Median :  4.9444        Median :    4.65          Median :   5.0
##  Mean   :  9.5313        Mean   :  138.22          Mean   : 102.6
##  3rd Qu.: 10.0294        3rd Qu.:   39.40          3rd Qu.:  54.5
##  Max.   :392.1569        Max.   :67115.00          Max.   :3300.0
##  NA's   :14129           NA's   :11606             NA's   :14342
##  MobilePhonesOwned  WillingToPayForConnection
##  Min.   :       0   Min.   :1.000
##  1st Qu.:       1   1st Qu.:1.000
##  Median :       1   Median :2.000
##  Mean   :   24392   Mean   :1.533
##  3rd Qu.:       2   3rd Qu.:2.000
##  Max.   :97985235   Max.   :4.000
##  NA's   :7875       NA's   :10680
##  EstimateOfElectricitySpendingLocal EstimateOfElectricitySpendingUSD
##  Min.   :    0                      Min.   :  0.0000
##  1st Qu.:    2                      1st Qu.:  0.0212
##  Median :  250                      Median :  1.9608
##  Mean   : 1316                      Mean   :  4.9201
##  3rd Qu.: 1000                      3rd Qu.:  4.9020
##  Max.   :50000                      Max.   :490.1961
##  NA's   :14734                      NA's   :14734
```

Dimension

Our dataset has 19449 rows and 25 columns. It is in long format, and each row represents a household.

Error

From the summary function, we can see that the grid connection fee, internal wiring fee, electricity bill price, electricity consumption, and outage duration all have a minimum value of -8. By investigating the dataset, all of these "-8" values come from Zambia, so we might need to go back to the original dataset to debug.

The "Grid connected" column for Liberia is a value ranging from 3-900 while the other countries have values of 1 or 2 where 1 means "yes" and 2 means "no". Need to verify if we used the correct columns from the Liberia dataset.

The "Outage duration hours" column for Liberia is also very large, with a max of 3300 hours. This number should be the outage duration hours in a week, and a week only has 168 hours. Need to go back to the Liberia dataset to check if the column is correct, and if the unit is in hours or minute.

"Electricity consumption kwh" column for Niger has a max of 67115.00. This column should be electricity consumption kwh per month. Niger in general has large numbers which is unreasonable for a household to use this much electricity. Therefore, might need to double check the unit or see if these are extreme values.

Formatting

The electrification status are supposed in numeric values representing different categories, however this column for Niger is "character" with a written description.

Extreme value

Dwelling room number has a max value of 2000 in Liberia which is unreasonable. Mobile phone number has a max value of 97985235 in Niger which is also unreasonable and need to be removed.

Missing Value

```
colSums(is.na(All_Country))
```

```
##                         Country                           HH_ID
##                               0                              17
##                      Subdivision1                      Subdivision2
##                               2                               2
##                      UrbanOrRural             ElectrificationStatus
##                              3297                           10811
##                     HouseholdSize                     DwellingType
##                              7119                             292
##                   DwellingOwnership               DwellingRoomsNumber
##                               289                            3925
##                       BankAccount                   GridConnected
##                               287                            3810
##              NumYearsGridConnected           GridConnectionFeeLocal
##                             12616                           14453
##               GridConnectionFeeUSD           InternalWiringFeeLocal
```

```
##                               14453                               16765
##               InternalWiringFeeUSD          ElectricityBillPriceLocal
##                               16765                               14129
##            ElectricityBillPriceUSD          ElectricityConsumptionkWh
##                               14129                               11606
##                OutageDurationHours                 MobilePhonesOwned
##                               14342                                7875
##          WillingToPayForConnection EstimateOfElectricitySpendingLocal
##                               10680                               14734
##   EstimateOfElectricitySpendingUSD
##                               14734
```

```
HH_ID_NA <- All_Country %>% filter(is.na(HH_ID))
#view(HH_ID_NA)
```

Our Dataset has lots of NAs especially for the columns asking about more details of electricity bills, connection fees and outages, etc.

There are 17 NAs in the HH_ID column. After viewing the rows with a missing household ID, 16 rows are from Liberia and 1 is from Niger. The other columns of these rows that has a missing household ID are also mostly NAs. We might consider removing these rows from the dataset because we can't be sure these are data from unique households.

Duplicate

```
Duplicate <- All_Country %>%
  filter(!is.na(HH_ID)) %>%
  add_count(Country, HH_ID, name = "count") %>%
  filter(count > 1) %>%
  arrange(Country, HH_ID)
#view(Duplicate)
```

We looked for duplicates in the HH_ID column, because each household ID **should** be different. However, lots of duplicates came up, and the household size and dwelling type are different, even with the same household ID. Therefore the different rows with the same household ID should still be different households and we will keep these data.